Data Visualization Course Project
Process Book

Delay Information in the United States Airports and Airlines

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(alphabetical order)

Introduction

How Did the Idea Start?

Our team of three started our Data Visualization course project with first choosing a dataset that did not only allow us to provide **interesting visualizations**, but also enable us to provide an exciting **data story** which is insightful and useful for various stakeholders. We decided to use the dataset of the **flight delay information** in **US airports and airlines** from 2009 to 2018 (link), given the increasing attention to causes behind delays happening in different airports.

When preparing the first milestone, we found prior works exploring the dataset to some extent, mainly focusing on highlighting key **performance indicators** related to flight operations, or data **statistics** and **static** visualizations. However, we found that prior work lacks focusing on **interactive** visualizations designed to show **trends** over time. Thus, we came up with ideas such as showing **color-coded** delay information of airports and routes over the map of the US for **easy comparison**, as well as enabling passengers and flight stakeholders to **compare** and **extract trends** over time.

Another decision made by our team in our initial exploration phase was to include the **decomposed delay information**, comparing the **different causes** of delays. We got the inspiration for this feature after conducting our **exploratory data analysis**, which showed us the set of features obtainable from the dataset. This comparison allowed us to form a better **storyline** around our visualizations, and increased the **relevancy** of our visualizations for the target audience of air travel stakeholders. We provided our **data story** next to our visualizations in the website, assisting visitors to explore our visualizations accompanied by certain insights that can deepen their engagement.

Pre-processing of Data

We conducted pre-processing on our data by pre-computing necessary **aggregations** for our visualizations, as the **large size** (approx. **8 GB**) of the raw data did not allow real-time computation for rendering the plots. Moreover, we found no data was available in the dataset for some periods; to ensure consistent patterns in the plots when shown over time and avoid sudden "drops" in values to zero when following the patterns over the time axis, we completed the data using **linear interpolations** by taking the mean of the period before and after, per each value.

Visualizations

Based on our story, as well as the features from the dataset investigated from our exploratory data analysis, we decided on implementing the visualizations below.

Interactive US Map with Delay Data

An interactive map of the US with **zoom control** and **time slider**. Each airport is indicated by a **circle**, with the **size** indicating the amount of flights and **color** indicating delay. Clicking on each airport highlights **all routes** to other connected airports.

Goal: to explore and compare airports' delay information over time based on their location, plus the routes between airports, in order to get insights about various travel paths connected to each airport.

Delay Leaderboard

An interactive leaderboard showing the **top 10** airports or airlines having the highest or lowest average delay at any time period, based on the user's selection.

Goal: to find the best and worst performing airports and airlines at any given time, useful for stakeholders to connect this data with possible causes of changes in the leaderboard.

Comparison of Delay Reasons

An interactive bar plot, allowing users to add or remove airlines and airports to see and compare the **distribution** of the different **causes** of flight delays individually for each time period.

Goal: to explore the causes contributing to delay in different times and for different airlines and airports for stakeholders; compare various airlines and airports regarding their delay information for customers.

Delay Patterns and Performance Indicators

An interactive parallel-coordinates plot that draws a polyline for every airline or airport across ten **performance-related** axes (flights, delays, cancellation rates, etc.).

Goal: to identify the trends, patterns, and correlations in airline and airport performance by allowing users to filter and compare airlines and airports across selected metric ranges.

Route Visualization

An interactive visualization in which users input the name of airports and see a visualization of the **paths** in terms of a **circular layout graph (chord)** for different times.

Goal: to simultaneously identify route delays between different airports; useful for stakeholders to understand the network, optimizing operations in different seasons.

Implementation Details

After deciding on the visualizations, we started to implement our **website** (tested on Safari and Chrome Mac), having a first version with static visualizations for the second milestone and later completing it to a **full version** until the third milestone.

We first conducted design sketches of our visualizations in **pen and paper** and **digital stylus** during the brainstorming meetings of our team, before going on to implement them using the **D3** framework.

Main Tech Stack

D3

A framework which we learned about for this course, and then we used it to implement the interactive visualizations for our website. We also used the dataframe functionalities (e.g., the csv function) of D3 to help with loading data.

Vite

A modern and simple framework for fast development and packaging of our web application. It was useful for us due to the fast live updates on each file save, facilitating the development process of the website.

React

We used React due to prior familiarity and the ability to use different modules/components to facilitate development in the team with best practices. We used TypeScript for type safety and better code maintainability.

Python

We used Python for the initial data analysis in milestone 1. Moreover, we conducted the preprocessing steps (see page 2) of the data using Python, to only load smaller data files for the final website in the end.

GitHub Pages

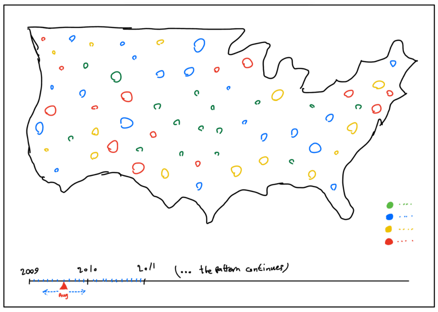
We hosted our website on the GitHub Pages platform. This allowed us to have a website supported by the GitHub infrastructure that can permanently stay online, reducing the need for us to maintain and pay for a server in the long term.

To follow **best practices** in coding and **reduce duplication**, we used **reusable components** throughout our React web application. For example, we used a reusable **time slider** component, featuring a **play button** which will automatically start the time slider progressing from the beginning to the end, enabling users to see the patterns in the visualizations evolving over time without the need to manually move the slider. As another example, we used an **airport/airline selection** component, featuring a **search textbox** for easily finding the names of the airports/airlines using their code or full name.

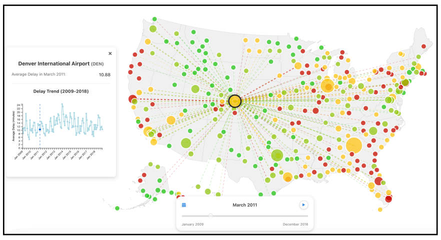
Explanation of the Main Visualizations

Interactive US Map with Delay Data

This component is the "main" interactive visualization of the website, encompassing a summary of our project, thus it was put as the first visualization in our developed website. We aimed to include two main features into the map: **A) airport-based** information, namely the delay and flight count information per airport which enables users to compare airports, and **B) route-based** information, which enables users to compare routes connected to a single selected airport.



From sketch to implementation!

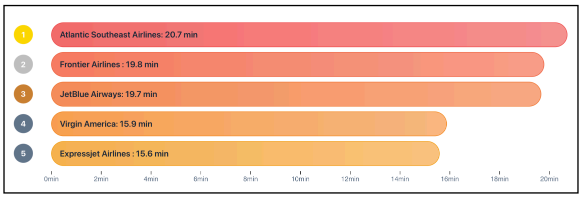


In our team, we decided that the best way of visualizing both A and B in the same plot is first showing only A to the user - and only showing B in the case user requests for it for a certain airport - in order to reduce user interface clutter and follow the human-computer interaction principle of **detail on demand** discussed in the paper "Dashboard Design Patterns" by [Bach et al. (2022)](). To implement this, we showed each airport as a **circle**, with the **radius** of the circle proportional to the number of flights passing through the airport.
We then **colored** the circles by the four colors of dark green, light green, yellow, and red (slight change from our early sketch), based on the quartile of total delay values they belonged to.

We show more details (part B) when the user clicks on each circle. Particularly, we added a **card** view, which was our **new idea** compared to our sketch in the previous milestone. The card shows the full name and the code of the airport, as well as the average delay in the currently selected time period. Moreover, a **plot** of the average delay of the airport over time is also shown, allowing the users to see how the delay data in any specific airport change over time. A **vertical bar** in the mini-plot shows the position of the current time period, with real-time changes reflecting the value of the slider below the page. Moreover, the **connections** to the other airports from the selected airport are also shown, with the color representing delay and the thickness representing the count of flights. This enables users to find the well-connected airports. **Hovering** on each circle shows the code and name of the airport to find the name of the connected airports easily.

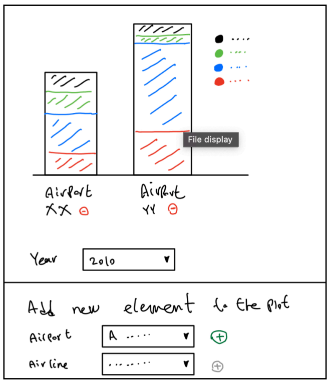
Delay Leaderboard

This component is a simpler yet useful interactive visualization, showing the **top 10** airlines or airports regarding delay information. We included a **segmented control** with a design generally following the best practices provided by [Apple](), enabling users to select whether they want to see information on airlines or airports. A button on the top right allows users to change the **direction**, seeing the airports or airlines with most or least delays. We used a **color mapping** of shades of green and red to indicate the delay value, while still providing the value per minutes in the end of each bar. On the left side, we included the **rank** of the airline, with the medal colors (🥇 🥈 🥉) for the top three. We put the name inside the bars whenever possible, and moved them after the end of the bar in case the bar was too small to fit the name. This visualization did not exist in our milestone 2 sketches and we decided to add it for milestone 3.



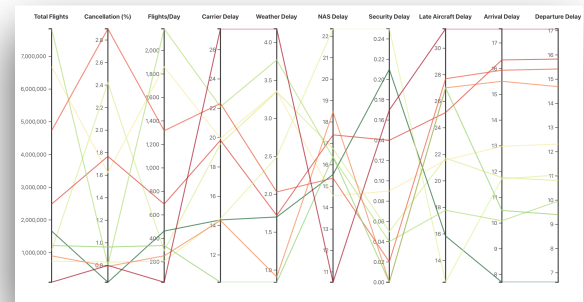
Comparison of Delay Reasons

While the leaderboard plot allows the users to compare delay of the best or worst airlines and airports, it doesn't allow the users to compare certain **selections**. As a result, we implemented this plot, including a component for airline or airport selection functionality to be able to add or remove selections. The plot compares the airlines and airports, as well as their **distribution** of **delay causes** (weather, carrier, etc.) at each time period. We opted for a more diverse color palette as opposed to different shades of the same color, for better **visibility** and **separation** of colors inside each bar. We generally used the same idea as mentioned in milestone 2, but replaced the dropdown element addition component with a **list view**, coming with a more intuitive design and a higher ease of use.



Delay Patterns and Performance Indicators

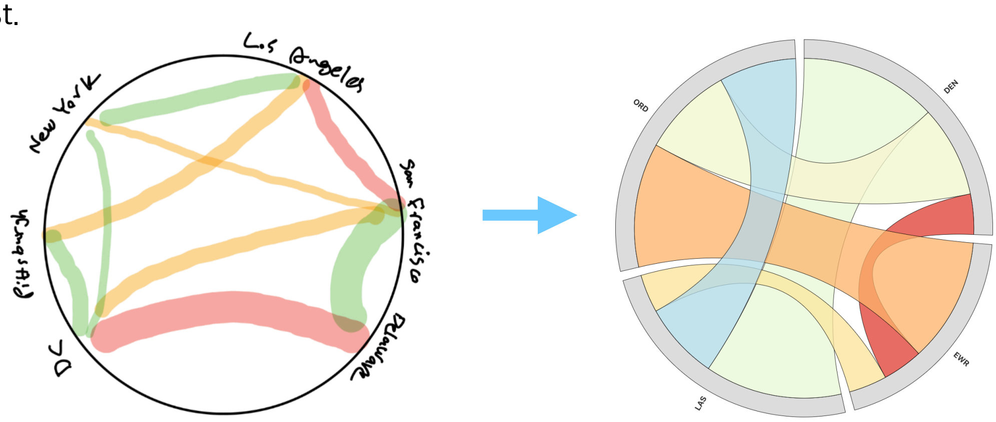
The parallel coordinates visualization offers an interactive way to explore and analyze airline



performance data. Each airline is represented as a **polyline**, colored based on the average delay, that traverses a set of parallel axes, where each axis corresponds to a specific **performance metric** such as 'Total Flights', 'Cancellation Rate', 'Average Delay', and various subcategories of delays. It enables users to discover **relationships** and **correlations** between different performance indicators. Users can interact with the plot by **dragging** along one or more axes to select specific ranges for the metrics, dynamically filtering only those airlines falling within the selected criteria on all dragged axes. Hovering over lines reveals **tooltips** with airline-specific information. This visualization did not exist in our milestone 2 sketches and we decided to add it for milestone 3.

Route Visualization

An interactive visualization in which the users can select different airports and see the routes between them visualized using a **circular layout graph** (**chord**). This plot allows the users to be **focused on certain airports**, reducing the clutter needed to investigate the routes compared to the US map visualization by a significant amount. Routes are **color coded** with delay information, and the **thickness** indicates the count of flights. Moreover, the length of the **airport** in the **perimeter** of the circle indicates the number of flights from / to that airport. Users can see the information of each particular route by **hovering** their mouse over the corresponding route, particularly the number of flights and the average delay of the route. The **time slider** on the bottom allows the users to see how certain routes evolve over time regarding delay information. While this visualization uses the overall same idea as mentioned in our milestone 2, we had initially planned to allow users to enter **city** names instead of **airport** names, to show the best airports to use for each city. However, we found very limited cities in the US with more than two relevant airports. As a result, we opted for providing the selection of airports instead of cities. Moreover, we added the **city name** to the **search** functionality in the list views, so that, for example, searching New York will lead to the JFK airport be shown in the list.



Challenges

We did not face major issues due to the previous experience of the team in web development, but we faced challenges unique to this project and the dataset, some of them included:

- **Large dataset:** the large amount of dataset made it impossible to work with on-the-fly, with even simple analyses in our early milestones taking **multiple seconds** to finish running. To solve this, we **pre-computer aggregated data** to use in the interactive visualizations.
- **Out of range coordinates:** a certain bug that took us a long time to fix was that some airports caused the map to **fail to render**. We initially did not find any pattern in the "failing" airports (we checked for Unicode characters in the names, wrongly-formatted IATA, null values, etc.), but then we found that those airports belong to the US territories which **do not belong** on the main map of the US (particularly the states PR, VI, AS, GU, MP). We then **removed** those states from the plot data to solve the bug.
- **Details of user interface:** we occasionally encountered challenges in certain website elements. For example, we initially put all the names **inside** the bars of the **leaderboard**, only to find out later that many names **did not fit** the bars when the corresponding value was small. We solved it by moving the text **after** the end of the bar when the length of the bar was less than six times the text length, a value found by trying and testing. Moreover, we faced difficulties in showing **multiline** texts in the **tooltips**, with text **overflowing** the tooltip frame. To fix this, we used the "measureText" function to check the current rendered width, and created a new line in a list manually whenever the current line overflowed the width.

Peer Assessment

All team members participated in most parts of the projects, with the majority of implementation and design decisions happening in meetings (either over Google Meet or in person on campus). With that said, some team members were focused more on certain aspects of developing the website, process book, screencast, or other deliverables throughout the project:

👨‍✈️ Aryan Ahadinia

Set up the main basis of the website. Main maintainer of the code and web app. Developed multiple components (e.g., chord diagram, parts of leaderboard, and the first version of the map). Conducted code cleanup and organization. Developed reusable components. Refined the process book. Participated in writing the README and debugging the visualizations.

📊 Matin Ansaripour

Worked on initial and subsequent pre-processing of the dataset obtained from Kaggle. Developed the parallel-coordinates visualization. Involvement in brainstorming and idea refinement. Conducted the related work search and most of the EDA. Refined the process book. Participated in writing the README and debugging the visualizations.

🚀 Seyed Parsa Neshaei

Co-developed and debugged multiple components (e.g., leaderboard, delay comparison, and route visualization in the map). Designed the initial sketches and mockups. Main maintainer of the milestone reports. Wrote the process book. Recorded the screencast. Wrote the data story for the website. Participated in writing the README and debugging the visualizations.

May our visual insights offer a clearer understanding of the patterns behind delays, and maybe, even a few ideas for a smoother flight journey ahead!